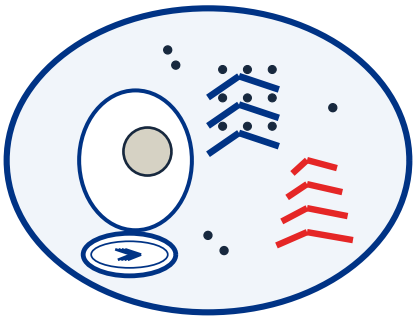
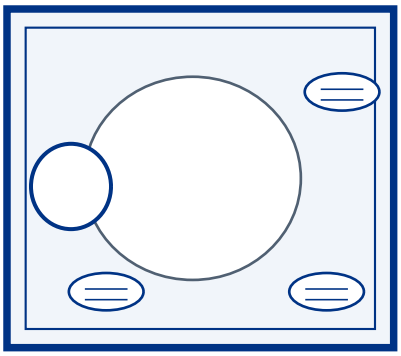
# Three cell types compared

Generalized animal, photosynthetic plant and bacterial cells: shared and distinct parts.

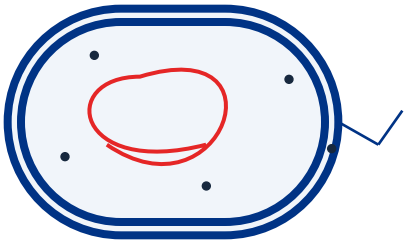
**Animal**



**Photosynthetic plant**



**Bacterium**



| Feature | Animal | Plant | Typical bacterium |
|---|---|---|---|
| Plasma membrane | Yes | Yes | Yes |
| DNA + ribosomes | Yes | Yes | Yes |
| Membrane-bound nucleus | Yes | Yes | No; DNA in nucleoid |
| Cell wall | No | Cellulose-rich | Usually peptidoglycan |
| Chloroplasts | No | In photosynthetic cells | No chloroplasts |
| Large central vacuole | No typical central vacuole | Often prominent | No equivalent vacuole |

Exceptions matter: a root cell may lack chloroplasts; absence of chloroplasts does not prove animal identity. Some bacteria lack walls. Archaeal walls differ from typical bacterial peptidoglycan walls. Cell shapes and sizes here are schematic.

# Three cell types compared

Self-check questions and explanations. Refer to the visual guide on the front.

### What distinguishes the three generalized cell plans?

Animal and plant cells have nuclei; the bacterium has a nucleoid. The photosynthetic plant cell has chloroplasts, a wall and a large central vacuole. The typical bacterium has a chemically different wall.

### Which major structures are shared by these cell types?

All three generalized living cell types have plasma membranes, ribosomes and DNA. Nuclei are eukaryotic; chloroplasts occur in photosynthetic eukaryotes such as many plant cells, not bacteria.

### What does a large plant central vacuole do?

A central vacuole can store substances and contribute to turgor/volume regulation. Its tonoplast is a membrane inside the cell; the wall is outside the plasma membrane.

### Does a nucleus with no chloroplasts prove animal identity?

A membrane-bound nucleus rules out a typical bacterium. No chloroplasts does not prove animal identity: fungi and many protists lack them, and some plant cells do too.

### Why can a root cell lack chloroplasts?

Different plant tissues have different functions; roots often lack chloroplasts. A cellulose-containing wall, large vacuole or other tissue/organism evidence can support plant identity. Do not rely on rectangular shape alone.

### How do typical bacterial and plant walls differ?

Plant walls are cellulose-rich with other polymers. Typical bacterial walls contain peptidoglycan, a chemically different network. Some bacteria lack walls; archaeal walls differ from typical bacterial walls.